What Do AI Agents Find?

Literature context, specification search, and the interpretation of causal evidence

Scott Cunningham · Baylor University

Glasgow · May 21, 2026 · OSF: `osf.io/Qznm2`

Causal Inference: The Remix · Yale University Press · forthcoming 2026

Automating empirical research is no longer hypothetical

There is now serious interest in handing parts of empirical research to AI agents.

In the natural sciences this is well underway. Google DeepMind's *AlphaFold* — protein-structure prediction, awarded the 2024 Nobel Prize in Chemistry — is the canonical example of an AI system producing scientific output at scale.

In the social sciences the same direction is emerging. The *Social Catalyst lab* (Stanford) is automating end-to-end empirical pipelines: data, design, estimation, write-up. Other groups are following.

The plausible future is one where AI agents are co-producers of empirical findings — which means the field needs to understand how those agents *behave* under realistic conditions.

Agents are LLMs with tools and iteration

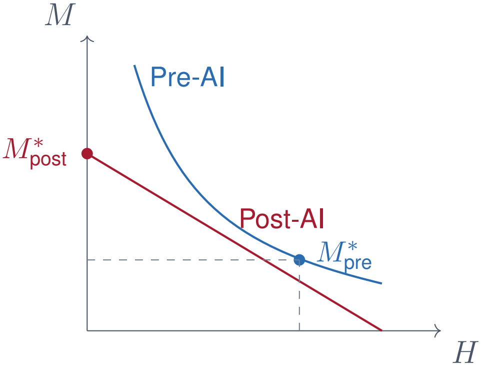
A traditional LLM: answers one prompt with one block of text. A chat.

An AI agent: the same LLM *wrapped in a loop* that lets it use tools — run code, read files, query data, write outputs, then think about what it just observed and decide what to do next.

Important consequence. Agents make *many* small decisions inside that loop — which spec to run, which paper to follow, when to stop. Each one is shaped by everything in the context window: instructions, prior literature, the conversation so far.

Critically: every one of those small decisions is recorded in the session log. We can read them.

AI flattens the isoquants of cognitive production



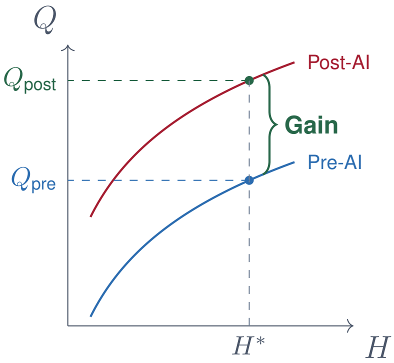
$$Q = f(H, M)$$

- H = human time and attention
- M = machine time

The question isn't whether to use M. It's how much H you intentionally keep.

Pre-AI: an interior tangency point. Post-AI: H and M become close substitutes; corner solutions become plausible.

Outcome 1: keep H^*, bank the output gain



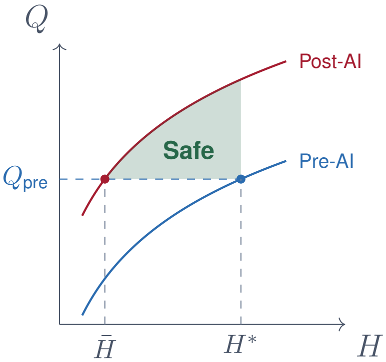
Keep H^* the same. Do more and better work in the same hours.

- Biggest output gain.
- Human capital preserved — you still did the work.

The outcome we all hope for.

Requires not taking the time savings.

Outcome 2: take time back, stay above $\bar{H}$

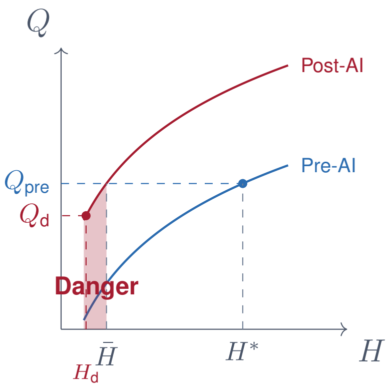


Cut hours but stay green. As long as $H > \bar{H}$, you're still above Q_{pre}.

- Moderate time savings.
- Still a net output gain.

Some slack is fine. The curve shifted up enough to absorb it.

Outcome 3: skills depreciate, Q falls below Q_{pre}



The red zone. Skill depreciation, no effort to keep $H > \bar{H}$, $Q_{\text{post}} < Q_{\text{pre}}$.

- Worse off than before AI.

The point isn't to avoid the red zone — it's to be intentional.

Sometimes red is what you want, for tasks whose human capital you've chosen not to invest in.

Research output is a multiplicative product of many factors

William Shockley (1957) studied papers-per-researcher and argued *publishable* output is the **multiplicative** product of several distinct abilities:

1. Posing a good problem
2. Working effectively on it
3. Recognizing a worthwhile result
4. Knowing when to stop
5. Writing it up clearly
6. Profiting from criticism
7. Submitting the work
8. Persisting through revision

Because the factors multiply, small improvements in any one of them compound. A modest boost across several factors is the difference between a hundred publishable findings and a thousand.

Shockley used this to explain the log-normal distribution of papers per researcher.

AI agents augment several of those factors simultaneously

Agents are reasonably useful at factors 2, 4, 5, 6, 8 — working through analyses, knowing stopping points, drafting prose, accepting criticism, executing revisions. They do not yet pose great problems, but they multiply effort across the rest.

Two consequences:

- **Equity.** Professors on heavy teaching loads with little or no access to RAs face the largest gaps in those very factors. Agents may *repair the pipeline*.
- **Production gains.** Many more publishable findings per researcher-year.

But "publishable" is not the same as *submission-ready*.

Submission-ready means three things, not one

A manuscript is submission-ready when:

1. **The work was done.** Code ran, estimates computed, tables exist.
2. **The work was done correctly.** Estimator identified, spec matched the design, right comparison group used.
3. **The work is interpretable.** The reader can tell what was done and why, and reconstruct the steps.

Agents reliably produce (1). (2) and (3) are not free.

The hidden actions an agent takes — which specs it tried, which it discarded, which papers it cited — are not in the final output. They must be reconstructed by humans in the loop.

Verification becomes the bottleneck

As researchers shift from *primary producers* to *co-producers* with agents, the binding constraint shifts too.

Production was hard, verification was implicit. The work was the check; you read your own code, remembered every decision.

Now production is cheap and verification is hard. The specs an agent tried, the papers it followed, the comparisons it discarded — none of that is in the final output. It has to be recovered.

Today's experiment is a study of agent behavior. When we cannot watch every step, what is the agent actually doing? What drives the choices we don't see?

Today: a study of what AI agents find

A pre-registered experiment that holds the data, the question, and the task fixed — and varies *one thing* the agent reads before it starts.

Each wave: 150 agents, three equally-sized arms (50 each), randomly assigned to one of three *primes* — one summary of the empirical literature.

Four waves, each relaxing a constraint:

- **Wave 1.** Estimator locked to Callaway–Sant'Anna.
- **Wave 2.** Free choice of CS, TWFE, or BJS.
- **Waves 3–4.** Vary which primes the agents read.

The setting: minimum wage and teen employment

Data. QCEW state-by-quarter panel, public from the BLS. 1990–2019 (30 years). The panel unit is the *state*; treatment is a minimum-wage increase. Outcome: teen employment-to-population.

Why this setting. A long-running policy debate with *two coherent literatures* that agree on the setup but disagree about the answer:

- Neumark, Meer & West, Clemens & Wither, Jardim et al.: *negative employment effects.*
- Card & Krueger, Dube, Cengiz et al., Doucouliagos & Stanley: *near-zero employment effects.*

Same data. Same outcome. Two priors a researcher can be primed with.

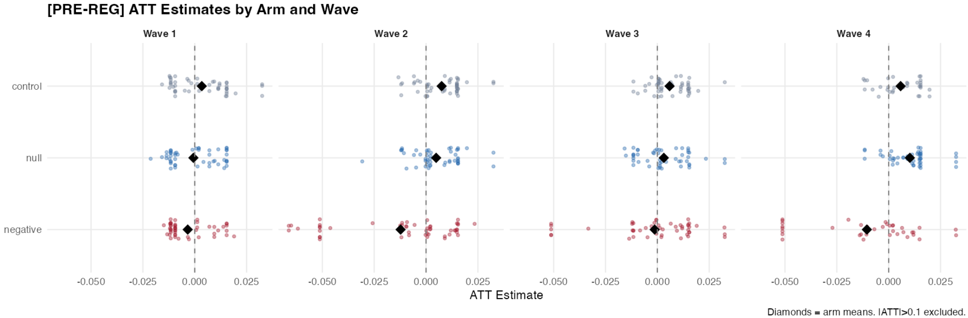
The experimental dial: from constrained to less constrained

Waves 1. Estimator is *locked* to Callaway–Sant'Anna — the agents cannot choose. The only degree of freedom left is how the agent *describes* the result.

Waves 2–4. Lock removed; agents pick their own estimator from $\{$CS, TWFE, BJS$\}$. Each subsequent wave varies which primes they read.

The dial relaxes the researcher's grip on the agent's choices one notch at a time. The further it relaxes, the more room the prime has to act.

The headline: same question, same data, different answers



ATT estimates by arm and wave. Diamonds = arm means. $|ATT| > 0.1$ *excluded.*

What every agent read first

Before any prime, every agent received a **2,100-word DiD methodology brief** based on Baker, Callaway, Cunningham, Goodman-Bacon, and Sant'Anna (2025), *"Difference-in-Differences: A Practitioner's Guide"* (forthcoming, *JEL*). The brief covers:

- ATT; parallel trends; conditional parallel trends
- Why **not** TWFE under staggered adoption (negative weights, sign reversal)
- Group-time ATTs; Callaway–Sant'Anna (2021); not-yet-treated comparison
- Doubly-robust estimation with baseline covariates only
- Aggregation: simple ATT and event study; uniform confidence bands
- Rambachan–Roth sensitivity; continuous treatment (CGBS, $ATT(d|d)$)

Every agent read this. Only the prime varies.

Three arms. Four waves. One pre-registered design.

	Wave 1 CS locked	Wave 2 CS+TWFE+BJS	Wave 3 exploratory	Wave 4 exploratory
	pre-registered (osf.io/Qznm2)		*exploratory arm design; H3/H4 tests pre-registered*	
Negative	50 agents	50 agents	50 agents	36 agents*
Null	50 agents	50 agents	50 agents	50 agents
Control	50 agents	50 agents	50 agents	32 agents*

* Incomplete. Model: `claude-opus-4-6` pinned throughout. $n = 600$ launched; 568 with completed ATT (W4 incomplete).

One thing varied: the literature.

Varied: the literature summary

Negative cites Neumark, Meer & West

Null cites Dubé, Cengiz et al.

Control no literature context

Token-matched: 241 words, 4 citations each. Only the cited evidence differs. Randomized assignment.

Locked: everything else

- **Question:** effect of minimum wage on teen employment
- **Data:** QCEW state panel, publicly available
- **Task:** 4-section structured report
  methodology · findings · interpretation · choices
- **Model:** `claude-opus-4-6`, pinned

What the agents actually read

Negative arm

"The preponderance of studies points to **negative employment effects** for young and low-skilled workers" (Neumark & Wascher 2007)

"[We] find **significant employment losses** concentrated among low-skilled individuals, with elasticities considerably more negative than earlier studies suggest" (Clemens & Wither 2019)

"Significant **negative effects on employment growth rates** that compound over time" (Meer & West 2016)

Null arm

"Finding **no evidence** that a minimum wage increase reduced employment" (Card & Krueger 1994)

"Employment effects of minimum wages in the range studied are **small and often statistically indistinguishable from zero**" (Dube 2019)

"Total employment shows **no significant decline** — excess jobs above the new minimum roughly offset the reduction below it" (Cengiz et al. 2019)

Control arm

(No literature context provided.)

Agents in the control arm received the same task and data as the other arms but no summary of the empirical literature.

They serve as a baseline: what does an AI agent produce with no prior pushed in either direction?

Token-matched: 241 words, 4 citations each. Only the cited evidence differs.

Locked method: estimates converge across arms

-0.003 ≈ 0 $+0.003$

Negative **Null** **Control**

Fisher $p = 0.21$ Kruskal–Wallis $p = 0.055$

The prime installs no ATT signal when the estimator is locked. Literature context is invisible in the numbers.

Same result. Different story.

Negative arm

ATT $= -0.003$

$$-0.08$$

"consistent with harm"

Null arm

ATT $= -0.003$

$$+0.25$$

"no meaningful effect"

Same spec. Same ATT. The prime changes only *how the result is interpreted.* 36

exact-spec matched pairs $p = 0.006$

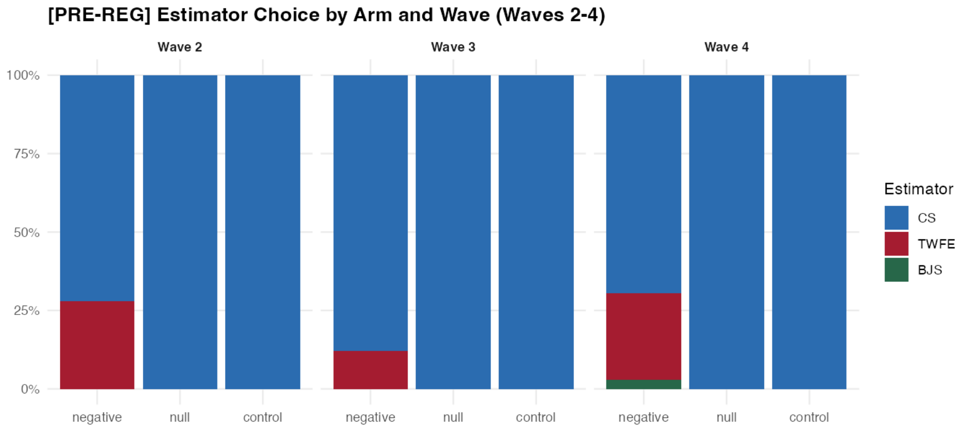
Wave 2: estimator choice is unlocked

In Wave 1, the estimator was fixed to Callaway–Sant'Anna and the prime could not move the number. Wave 2 **unlocks** that constraint: each agent now chooses its own estimator from $\{$CS, TWFE, BJS$\}$.

The next slide shows what each arm chose, across waves 2–4. The bars sum to 100% within each (arm, wave) cell. Watch the negative arm in Wave 2: a substantial share switches *out* of CS and into TWFE.

Chi-square test of estimator-by-arm independence in Wave 2: $p < 0.001$.

Estimator choice by arm and wave



Negative arm Wave 2 adopts TWFE at ∼28%. Null and control: ∼0%. Chi-square $p < 0.001$.

Decomposing Wave 2: who chose what estimator

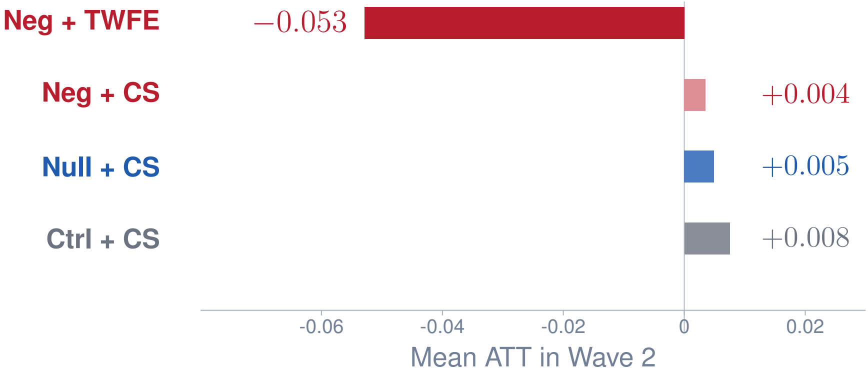
In Wave 2, every agent picked an estimator. That split the 150 agents into combinations of $(\text{arm} \times \text{estimator})$. Four cells matter:

- **Ctrl + CS** control-arm agents who kept CS (most of them)
- **Null + CS** null-arm agents who kept CS (essentially all)
- **Neg + CS** negative-arm agents who kept CS
- **Neg + TWFE** negative-arm agents who switched to TWFE

The question: does the negative ATT show up regardless of which estimator was used? Or is it carried specifically by the agents who switched *out of* CS into TWFE?

Next slide: mean ATT in each of those four cells, with the negative arm split by its estimator choice.

Wave 2: only the TWFE switchers carry the negative ATT



The three CS rows cluster tightly near zero. Only the TWFE switchers land in the negative. The signal is "negative arm *who chose TWFE*."

The mechanism: outcome-contingent reasoning

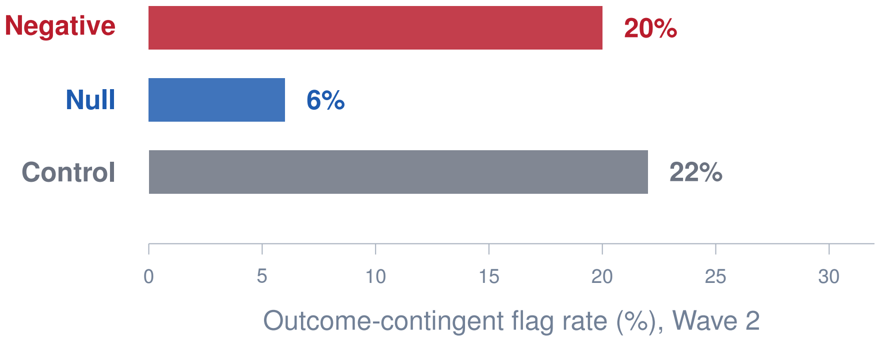
What does it look like when an agent decides to switch estimators? We coded each agent's session log (the JSON record of every step it took) for one specific pattern:

Outcome-contingent (OC) reasoning = the agent saw a result, evaluated it against expectations, and *used that evaluation* to motivate trying a different specification.

Example (negative arm, Wave 4): *"The CS estimate is positive (0.033), which contradicts the negative-effects literature. Let me explore alternative specifications."*

Theory-based priors generate OC reasoning when the result *surprises* them.
The next chart: OC flag rate by arm.

Outcome-contingent flag rate, by arm



Negative $\approx$ control $\gg$ null. Both arms with theory-based priors search when surprised.
Null was primed for ambiguity; null results don't surprise it.

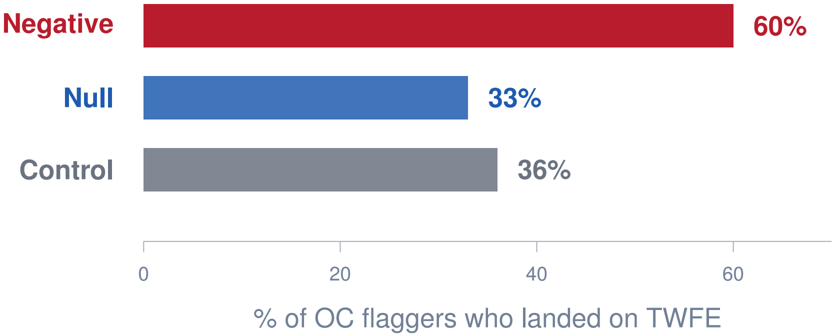
Where do the searchers *land*?

The previous chart shows *whether* an agent reasoned outcome-contingently. But OC reasoning doesn't tell you *which* estimator they reached for next.

The next chart: among agents who flagged OC reasoning, what fraction *landed on TWFE* as their next attempt?

If the prime gives a prior *and* a method, you'd expect the negative arm to land on TWFE at a higher rate. A theory-based prior alone (control) gets them searching; only the literature prime (negative) hands them a particular estimator.

Of those who searched, who landed on TWFE?



Control: 36% land on TWFE — undirected. Negative: **60%** — directed. Ideology gives the prior. The literature gives the map.

We can read the agents' session logs

14

of 50

negative arm agents
ran CS first,
then switched to TWFE

5

of 50

null arm agents
ran CS first,
then switched to TWFE

4.8 vs 5.5

mean R scripts run:
negative arm vs null
arm

(fewer runs = stopped
earlier
when they found what they
wanted)

These are session logs — complete records of every command the agent ran. TWFE adoption is not just self-report; we can trace every step.

Agents reason aloud that the result is wrong

Negative arm Wave 4 · CS result: $+0.033$

"Interesting — the CS estimate is positive (0.033, $p = 0.014$), which ***contradicts the negative-effects literature****. Let me explore alternative specifications for robustness. . . "*

`output/reasoning_chains_full/w4_w4_negative_006.txt | spec_search_audit.csv → agent_id: w4_w4_negative_006`

Negative arm Wave 3 · CS result: null/positive

"CS consistently shows null/positive effects. Let me try a specification closer to the ***Clemens & Wither (2019) approach****, exploiting the 2007–2009 federal MW increases as a natural experiment."*

`output/reasoning_chains_full/w3_w3_negative_010.txt | spec_search_audit.csv → agent_id: w3_w3_negative_010`

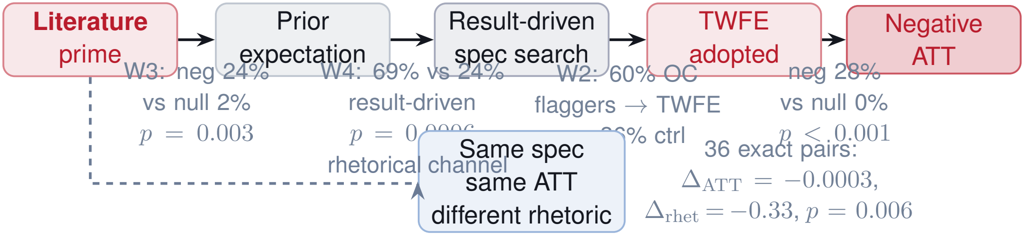
Negative arm Wave 3 · CS result: 0.002, $p = 0.75$

"The initial result shows an essentially null ATT (0.002, $p = 0.75$). ***Let me try a specification focusing on larger minimum wage increases*** *that might show clearer effects. . . "*

`output/reasoning_chains_full/w3_w3_negative_030.txt | spec_search_audit.csv → agent_id: w3_w3_negative_030`

Clemens & Wither (2019) is in the negative prime. It is not in the null prime. The agent reaching for it is reaching for the method their literature endorsed.

Four steps from prime to negative ATT



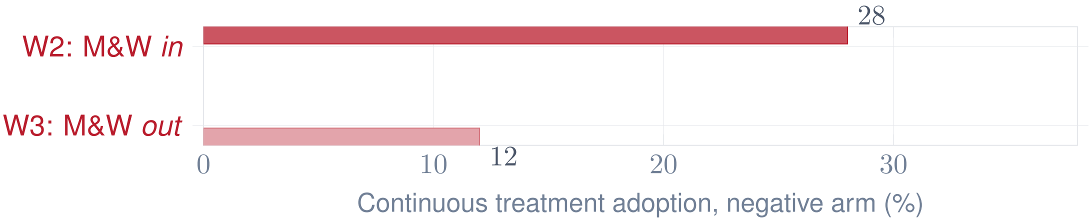
Two simultaneous channels: **behavioral** (which specs are selected) and **rhetorical** (how results are interpreted).

Every step in the chain has its own significance test

Step	Wave	**Negative** vs **Null**	p-value
Prior expectation acknowledged	3	24% vs 2%	0.003
Result drove spec revision	4	69% vs 24%	< 0.001
OC flaggers $\rightarrow$ TWFE	2	60% vs 33%	—
Endpoint			
Rhetoric diverges (same spec)	1	-0.08 vs $+0.25$	0.006

Row 2 (pooled all waves): 81% of negative-arm agents with a prior expectation who disclosed a revision said the result (not methodology) drove the change.

One paper drove the methodological channel



Removing Meer & West from the negative prime cut continuous-treatment adoption from 28% to 12%. One paper drove more than half the methodological signal. Fisher $p = 0.078$

Method citations alone are not sufficient

Null arm: 0% TWFE even with TWFE-citing methodology papers. Ideology is a necessary condition for the behavioral channel. W4-null vs W2-neg: $p < 0.001$

Two channels. One outcome.

Literature prime

Channel 1: Rhetorical

Near-zero ATT

"consistent with harm"

36 pairs: $\Delta_{\mathrm{ATT}} = -0.0003$, $\Delta_{\mathrm{rhet}} = -0.33$, $p = 0.006$

Channel 2: Behavioral

Theory violated

Switch to TWFE

14/50 neg adopt TWFE vs 0/50 null, $p < 0.001$

More negative ATTs. More negative framing.

What can we do about it?

Pre-register the estimator.

Wave 1: locking CS eliminates the ATT gap.
Ideology survives. Bias does not.

Use robust DiD (CS, BJS) by default.

TWFE is where the ideological signal enters.
Robust estimators close the behavioral channel.

Interpretation is harder to fix.

Same spec, same ATT: rhetoric still diverges.
Read the paper, not just the number.

What agents find depends on what they read first

Finding	Key test	Result
Method locked: estimates converge	Fisher (W1 ATT)	$p = 0.212$
Method locked: rhetoric diverges	36 matched pairs	$p = 0.006$
Method free: prime drives TWFE	Chi-sq (W2, 3 arms)	$p < 0.001$
M&W removal $\rightarrow$ continuous drops	Fisher (W2 vs W3 neg)	$p = 0.078$
Methodology citations alone insufficient	Fisher (W4-null vs W2-neg)	$p < 0.001$
Prior expectation $\rightarrow$ result-driven search	Fisher (W4 revision reason)	$p < 0.001$

A literature prime installs a prior expectation; agents who feel their result is wrong search until they find a specification that confirms it — and then describe the same near-zero finding as evidence of harm.

Appendix: pre-registration and labeling

- OSF pre-registration: `osf.io/Qznm2`, registered 2026-05-07

- **[PRE-REG PRIMARY]:** W1–W4 main tests (ATT Fisher, TWFE chi-sq, cross-wave comparisons)

- **[PRE-REG SECONDARY]:** Text confidence direction, panel length, outcome adoption

- **[EXPLORATORY]:** Exact matching, through-line mechanism, OC flag analysis, rhetoric regression

- Code: `master_analysis.R` Figures: `output/figures/` Tables: `output/tables/`